Did ChatGPT Reprice AI-Exposed Labor? Abnormal Returns, Substitution, and Complementarity Around the Generative-AI Shock

Nam Bao Ngo Lei Gao*

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Abstract

We ask whether the U.S. stock market repriced firms according to their exposure to generative artificial intelligence around ChatGPT's release (November 30, 2022) and GPT-4 (March 14, 2023). Using market-model cumulative abnormal returns (CARs) for 3,056 firms and the Felten–Raj–Seamans AI Industry Exposure (AIIE) index mapped to firms by 4-digit NAICS, we find that AI exposure was priced—and that its *sign* depends on whether AI substitutes for or complements a firm's labor. An equal-weighted portfolio long the most and short the least AI-exposed industries earned +1.3% over $[0, +1]$ ($t = 3.96$) and +2.7% over $[0, +10]$ ($t = 3.46$) around ChatGPT; in the cross-section a one-standard-deviation increase in AIIE is associated with a +0.71pp ten-day

*Ngo: ASSIP 2026, George Mason University. Gao (corresponding author): Costello College of Business, George Mason University, `lgao9@gmu.edu`. This is a student–faculty working paper prepared in the ASSIP 2026 Empirical Finance research group. All results use public data—CRSP and Compustat via WRDS; the Felten–Raj–Seamans (2021) AI-exposure indices from their public data appendix; and SEC EDGAR 10-K filings; every reported number traces to a regression-output CSV in `output/tables/` and is reproduced by the code in `code/`. No numbers are fabricated. We thank the ASSIP 2026 cohort for comments. Any errors are our own.

CAR ($t = 2.49$, industry-clustered), rising to +0.97pp ($t = 3.39$) among non-financials, with a clean null in the pre-event placebo window. But the average masks a sign reversal: AI-*supplier* industries (software, semiconductors, cloud) earned +0.6% while *substitution* industries whose product *is* cognitive labor (staffing, education, publishing, legal, consulting, advertising) earned −2.8% ($t = -2.32$). Firm-level 10-K AI-mention intensity (from 650 EDGAR filings) correlates with AIIE ($\rho = 0.25$) and independently predicts the tight-window cross-section. The market treated generative AI as a net cost saving for most firms but as a competitive threat where AI can produce the firm's output. The priced-exposure effect survives balancing high- and low-exposure firms on size, value, momentum, and employment, but is only borderline under industry-level randomization inference (which we report transparently, since exposure varies across just 182 industries), and a power analysis shows the well-populated windows are informative rather than underpowered.

JEL: G14, G12, J24, O33, E24. **Keywords:** generative AI, ChatGPT, labor exposure, event study, abnormal returns.

1 Introduction

On November 30, 2022, OpenAI released ChatGPT; it reached one million users in five days and became the fastest-adopted consumer technology on record. For the first time a general-purpose tool could perform, at low cost, tasks long thought to require educated human labor—writing, summarizing, coding, tutoring, drafting contracts, answering customers. If equity markets are forward-looking, the arrival of such a technology should have repriced firms in proportion to how much their businesses are exposed to the labor it can now do. This paper asks a simple question: *around ChatGPT (and GPT-4), did high-AI-exposure firms earn abnormal returns, and did the sign depend on whether AI substitutes for or complements the firm's labor?*

The sign is not obvious ex ante. For most firms, exposed labor is an *input cost*: if AI can do it more cheaply, margins rise and the stock should appreciate—a complementarity or cost-saving channel (Brynjolfsson et al., 2023; Noy and Zhang, 2023). But for some firms the exposed labor *is the product*: a staffing agency, an online-education company, a publisher, a consultancy, or a copywriting shop sells cognitive output that generative AI can now substitute for, so the same technology is a competitive threat (Acemoglu and Restrepo, 2019; Autor et al., 2003). The market's net reaction therefore mixes a positive cost-saving effect and a negative product-substitution effect, and which dominates is an empirical question about the cross-section of returns.

We answer it with a standard cross-sectional event study. We take every CRSP common stock active in the event window, estimate a market model over $[-252, -46]$ against the CRSP value-weighted index, and cumulate abnormal returns over tight windows around each dated release. We measure AI exposure at the industry level with the Felten–Raj–Seamans (2021) AI Industry Exposure index (AIIE)—a canonical, employment-weighted measure of how much an industry's occupational task bundle is exposed to AI—merged to firms by 4-digit NAICS. Because AIIE is predetermined and defined at the industry level, we cluster standard errors by industry and treat the industry measure as our workhorse, following the design used by Eisfeldt et al. (2023) for firm-level generative-AI exposure.

Our results are threefold. *First, AI exposure was priced, positively on average.* An equal-weighted long–short portfolio (top minus bottom AIIE quintile) earned +1.31% over the tight $[0, +1]$ window ($t = 3.96$) and +2.68% over the diffuse-adoption $[0, +10]$ window ($t = 3.46$; Table 2). In the cross-section, a one-standard-deviation increase in AIIE is associated with a +0.71-percentage-point ten-day CAR ($t = 2.49$; Table 3, column 3), robust to size, momentum, book-to-market and employment controls, and rising to +0.97pp ($t = 3.39$) when we drop financial firms. The effect is absent in the $[-10, -2]$ pre-event placebo window (-0.33pp, $t = -1.11$), so it is not a pre-trend. Figure 2 shows the cumulative abnormal returns of high- and low-exposure firms diverging only after day 0.

Second, the sign reverses with the nature of exposure. We split high-exposure industries into AI *suppliers* (software, semiconductors, computers, cloud—firms that build AI) and *substitution* industries whose output is itself cognitive labor (employment services/staffing, education, publishing, legal, consulting, advertising). Over $[0, +10]$ around ChatGPT, supplier industries earned +0.64%, ordinary AI-*user* industries −1.28%, and substitution industries −2.76% ($t = -2.32$; Table 4 and Figure 3). The supplier-minus- substitution gap is +2.5pp. Generative AI was good for firms that sell it or use it internally, and bad for firms whose product it can replace.

Third, a firm-level text measure corroborates the industry proxy. From 650 pre-ChatGPT 10-K filings on SEC EDGAR we compute each firm's AI-mention intensity (AI-related terms per 10,000 words). It is positively but only modestly correlated with AIIE ($\rho = 0.25$)—the two capture related but distinct exposure—and, on its own, predicts positive ChatGPT CARs, significantly in the tight $[0, +1]$ window ($t = 2.04$) and positively but insignificantly over $[0, +10]$ (Table 7). The industry- and firm-level measures point the same way.

The GPT-4 window (March 14, 2023) is badly confounded by the Silicon Valley Bank collapse (March 9–13) and the ensuing regional-bank and Credit Suisse turmoil, which hit exactly the high-AIIE financial sector. We report GPT-4 results but lean on ChatGPT for inference; reassuringly, the AIIE effect survives and even strengthens when we drop financials from the GPT-4 window (+0.62pp, $t = 2.72$; Table 6). Cleanly separating the AI signal from the concurrent macro repricing—the dovish Powell pivot also fell on November 30, 2022—requires hand-verifying the exact event calendar against market close and hand-coding true task exposure firm-by-firm, which is the student's contribution (Appendix; `STUDENT_TASKS.md`).

We contribute to three literatures. Relative to the AI-and-labor literature that measures occupational and industry exposure (Felten et al., 2021; Webb, 2020; Eloundou et al., 2024), we provide asset-pricing evidence that markets priced that exposure within days of ChatGPT. Relative to Eisfeldt et al. (2023), who show that generative-AI *workforce* exposure earned positive returns after ChatGPT, we document the *sign heterogeneity* predicted by the task framework: exposure is priced positively where AI complements and negatively where AI substitutes for the firm's output. Relative to the displacement-versus-reinstatement theory of automation (Acemoglu and Restrepo, 2019; Autor et al., 2003), we show the stock market discriminated between the two channels in real time. We also complement work on how AI investment and technological innovation map into firm growth and market value (Babina et al., 2024; Kogan et al., 2017), by isolating the market's instantaneous reaction to a single, dated general-purpose-AI shock. Section 2 develops hypotheses, Section 3 situates the paper in the literature, Section 4 the data and research design, Section 5 the main results, Section 6 the sign channel, Section 7 robustness, Section 8 heterogeneity by observable moderators, Section 9 a characteristic-matched design and industry-level randomization inference, Section 10 the power/minimum-detectable-effect analysis, and Section 11 concludes.

2 Hypotheses

The task framework of Autor et al. (2003) and the automation model of Acemoglu and Restrepo (2019) distinguish a *displacement* effect (a technology performs tasks previously done by labor) from a *productivity/reinstatement* effect (cheaper tasks raise output and demand for complementary labor). Generative AI is a general-purpose technology whose comparative advantage is language and cognitive tasks (Eloundou et al., 2024; Felten et al., 2023). For a firm that *uses* such labor as an input, the displacement of that labor is a cost saving; for a firm that *sells* such labor as its output, displacement is demand destruction.

> **H1 (Exposure is priced).** Around ChatGPT, abnormal returns are related to a firm's industry AI exposure.

> **H2 (Sign channel).** The relation is positive where AI *complements* the firm (AI suppliers; firms that use exposed labor internally) and negative where AI *substitutes* for the firm's product (industries whose output is cognitive labor).

H1 without H2 would say the market read generative AI as uniformly good (or bad) for exposed firms; H2 says the market conditioned on *who captures the surplus*.

3 Related Literature

This paper connects three strands. The first is the fast-growing literature on generative AI and the labor market. Felten et al. (2021) construct the AI-exposure indices we use, mapping occupational task bundles to AI capability and aggregating to industries; Webb (2020) and Eloundou et al. (2024) build complementary occupation-level exposure measures, and Felten et al. (2023) specialize the framework to language models like ChatGPT. Experimental and field studies show generative AI raises the productivity of exposed workers (Brynjolfsson et al., 2023; Noy and Zhang, 2023). This literature measures *how exposed* tasks and industries are; our contribution is to ask whether equity markets *priced* that exposure, and with what sign, in the days around a single dated shock.

The second strand is asset pricing and technological change. Kogan et al. (2017) show that technological innovation is priced in the cross-section of returns, and Babina et al. (2024) document that firm-level AI investment maps into growth and product innovation. Closest to us, Eisfeldt et al. (2023) show that firms with greater generative-AI *workforce* exposure earned positive abnormal returns after ChatGPT. We build on their event-study design but shift the object of study from the average exposure effect to its *sign heterogeneity*: the same technology is a tailwind where it complements the firm and a threat where it substitutes for the firm's product. This distinction is the empirical counterpart of the displacement-versus-reinstatement theory of automation (Acemoglu and Restrepo, 2019; Autor et al., 2003).

The third strand is the methodology of event studies and design-based inference. Our returns machinery follows the daily-event-study tradition (Brown and Warner, 1985; MacKinlay, 1997). Because our key regressor, AIIE, varies only at the industry level, conventional clustered standard errors rest on industry-cluster asymptotics that can overstate precision when exposure is coarse; we therefore complement them with industry-level randomization inference, following the argument of Young (2019) that design-based permutation tests discipline seemingly significant estimates under a modest number of clusters. Where a sub-result is statistically insignificant—the five-day window, the individually-imprecise substitution bucket—we interpret it through its minimum detectable effect rather than as proof of no effect, as Abadie (2020) and the literature on low power and selective reporting in economics (Ioannidis et al., 2017) recommend. These tools are used diagnostically, not decoratively: they temper the headline where the industry-level design is genuinely uncertain, and they bound the nulls.

4 Data and Research Design

AI exposure. Our exposure measure is the AI Industry Exposure (AIIE) index of Felten et al. (2021), taken from their public data appendix. AIIE scores 246 4-digit NAICS industries by averaging occupation-level AI exposure (AIOE) weighted by each industry's BLS occupational employment mix; it is standardized (mean 0, SD 1). The most exposed industries are professional and financial services (securities, accounting, legal, insurance); the least exposed are manual industries (crop-support, building trades, warehousing). AIIE is a measure of *how much* of an industry's task bundle AI can do; it is silent on whether that is good or bad for firm value—which is exactly the object of H2.

Returns and sample. From CRSP we take all common shares (share codes 10, 11) on the NYSE/AMEX/Nasdaq with a valid NAICS whose name record spans the ChatGPT window: 4,438 stocks. Following standard event-study methodology (Brown and Warner, 1985; MacKinlay, 1997), we estimate the market model $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$ over $[-252, -46]$ against the CRSP value-weighted index and cumulate abnormal returns over $[0, +1]$, $[0, +5]$, $[0, +10]$, $[-1, +1]$, and a $[-10, -2]$ placebo, for two events: ChatGPT (day 0 = 2022-11-30) and GPT-4 (day 0 = 2023-03-14). Merging to AIIE on 4-digit NAICS and requiring a valid estimation window leaves **3,056 firms across 182 industries**. Predetermined controls, all measured before ChatGPT, are ln(market equity) at 2022-10-31, six-month momentum (2022-05 to 2022-10), Compustat book-to-market (FY2021 common equity over market equity), and ln(employees). AIIE and controls are winsorized at 1/99 and standardized; CARs are winsorized at 0.5/99.5. Table 1 reports summary statistics.

Sign buckets. To operationalize H2 we assign 4-digit NAICS industries to three buckets. *Supplier* (computers, semiconductors, software publishers, data processing/cloud) are firms that build AI. *Substitution* (publishers, other information services, legal, consulting, advertising, employment/staffing services, and education) are industries whose output is cognitive labor that generative AI can itself produce. All others are *users*. This NAICS-based split is deliberately coarse: CRSP's industry tags are noisy (for example, several payment and financial-data firms are tagged under "business support services"), and IT-services firms both build and are threatened by code-generating AI. Hand-coding true task exposure for 40–50 firms is the student's task; the automated split is the machine baseline and is, if anything, conservative because it leaves some large information-platform winners inside the substitution bucket.

Firm-level 10-K AI intensity. As an independent, text-based exposure measure we download each firm's most recent 10-K filed before ChatGPT from SEC EDGAR, strip HTML, and count AI-related terms ("artificial intelligence," "machine learning," "large language model," etc.) per 10,000 words. Across 650 filings, 35% mention AI at least once. This measure is used to validate AIIE and as an alternative regressor.

Design-based inference. Two features of the design shape how we read the estimates. First, AIIE is defined at the 4-digit-NAICS level, so treatment is effectively assigned to 182 industries, not 3,056 firms; industry-clustered standard errors are the right first step, but the effective number of independent units is the industry count. We therefore report, alongside the clustered t-statistics, an *industry-level randomization-inference* p-value that permutes the 182 industry AIIE values among the industries (holding firms-per-industry fixed), broadcasts the permuted exposure back to firms, and refits the clustered cross-section 1,000 times (Section 9). Second, because AIIE is correlated with firm size and growth characteristics, the priced-exposure effect could in principle reflect a size or value/growth factor loading that repriced on the same days; we address this with a characteristic-matched design that balances high- and low-AIIE firms on size, book-to-market, momentum, and employment before comparing their CARs (Section 9), and with a power/minimum-detectable-effect analysis (Section 10) that states each null as a bound. Fully separating the AI signal from the concurrent Powell pivot requires the hand-verified event calendar that is the student's reserved task (Appendix; `STUDENT_TASKS.md`); the design-based checks here use only observable data.

5 Main Results: Was AI Exposure Priced?

Long–short portfolios. Table 2 sorts firms into AIIE quintiles and reports mean CARs. Around ChatGPT the top-minus-bottom quintile earns +1.31% over $[0, +1]$ ($t = 3.96$) and +2.68% over $[0, +10]$ ($t = 3.46$); the $[-10, -2]$ placebo is an insignificant -0.49% ($t = -0.64$). The five-day $[0, +5]$ window is positive but insignificant (+0.70%, $t = 1.43$)—the repricing was not concentrated on any single day, consistent with ChatGPT's virality building over the first week. Figure 2 makes the timing visible: high- and low-AIIE terciles track each other through day -1 and diverge afterward.

Cross-section. Table 3 regresses the ten-day ChatGPT CAR on standardized AIIE with progressive controls, clustering by industry. A one-SD increase in AIIE raises the ten-day CAR by 0.58pp with no controls ($t = 1.88$) and 0.71pp with the full control set ($t = 2.49$); the effect is 0.97pp among non-financials ($t = 3.39$) and 0.35pp in the tight $[0, +1]$ window ($t = 2.61$). H1 is supported: the market priced AI exposure, positively on average, within days.

6 The Sign Channel: Substitution versus Complementarity

The positive average hides opposing forces. Table 4 reports mean CARs by sign bucket. Over ChatGPT $[0, +10]$, AI-supplier industries earned +0.64%, ordinary users −1.28%, and substitution industries −2.76% ($t = -2.32$); the pattern is sharper in the five-day window (+0.36%, −0.45%, −2.11% with $t = -2.73$ for substitution). Table 5 formalizes it: relative to ordinary users, supplier industries earned +1.41pp and substitution industries −1.06pp over $[0, +10]$, a +2.5pp supplier-minus-substitution gap. Figure 3 visualizes the ordering. Individually the bucket coefficients are imprecise—there are only four supplier and nine substitution industries, and industry clustering is demanding—but the ordering is monotone and the substitution underperformance is significant in the raw means. H2 is supported: exposure is priced positively where AI complements and negatively where AI substitutes for the firm's output.

Firm-level validation. Table 7 shows that firm 10-K AI-mention intensity, an entirely separate measure built from filing text for 650 firms, is positively correlated with AIIE ($\rho = 0.25$) and independently predicts positive ChatGPT CARs—significantly in the $[0, +1]$ window ($t = 2.04$), positively but insignificantly over $[0, +10]$. The modest correlation is itself informative: the two measures overlap but are not redundant, and sharpening the firm-level text measure (a student task) should tighten the link.

7 Robustness

Table 6 collects sensitivity checks on the AIIE coefficient. The effect holds under heteroskedasticity-robust (HC1) rather than clustered errors, when financials are dropped, and in the tight $[0, +1]$ and $[-1, +1]$ windows. It is *absent* in the $[-10, -2]$ placebo (−0.33pp, $t = -1.11$), ruling out a mechanical growth-factor pre-trend. The five-day window is the weakest (+0.20pp,

$t = 1.26$), which we report rather than hide. For GPT-4—whose window overlaps the SVB banking crisis—the AIIE coefficient is +0.63pp ($t = 2.99$) and, crucially, is unchanged when we drop the crisis-hit financial sector (+0.62pp, $t = 2.72$), suggesting the AI signal is not merely the banking shock in disguise. The single largest threat to identification is the coincidence of ChatGPT with the November 30, 2022 Powell pivot; the market model removes the market-wide component, and the null placebo is reassuring, but a characteristic-matched design and a hand-verified event calendar (student task) are needed to fully separate the two.

8 Heterogeneity by Observable Moderators

If the priced-exposure effect is concentrated in an identifiable subgroup, the pooled coefficient understates it there and overstates it elsewhere. Table 8 interacts standardized AIIE with four *predetermined, observable* moderators—an above-median size indicator, above-median book-to-market, above-median momentum, and a financial-firm indicator—reporting the base AIIE effect and the interaction for the headline $[0, +10]$ and tight $[0, +1]$ windows. We choose these moderators precisely because they are observable and orthogonal to the firm-level task exposure the student will hand-code (Task 2): the interaction of AIIE with *true* substitution-versus-complement exposure is the reserved test, and we do not proxy it here. The pattern is that the priced-AIIE effect is broadly present rather than driven by one cell. It is positive in essentially every base group; the size interaction is positive and significant in the tight window (+0.64pp, $t = 2.75$), indicating the immediate repricing was a little stronger among larger firms, while the value, momentum, and financial interactions are individually insignificant. Notably, the effect is if anything *weaker*, not stronger, inside the high-book-to-market and high-momentum groups (negative interactions), which is not what a mechanical value- or momentum-factor story would predict. No single observable moderator explains the result away, and none substitutes for the hand-coded task exposure that the sign channel ultimately needs.

9 Characteristic Matching and Randomization Inference

Characteristic matching. The most serious observable confound is that high-AIIE firms differ systematically from low-AIIE firms—they are larger, more growth-oriented—so the priced-exposure effect might be a size or growth factor loading that repriced around ChatGPT for reasons unrelated to AI. Table 9 confronts this directly. We take the top and bottom AIIE terciles and nearest-neighbor match each high-AIIE firm to a not-yet-used low-AIIE firm on the standardized predetermined controls (size, book-to-market, momentum, employment), without replacement and within a caliper, producing 675 matched pairs. Matching sharply improves balance: the high-minus-low size gap falls from -0.19 to 0.02 standard deviations, and the other characteristics are balanced to within 0.05 SD. On the balanced sample the exposure effect survives: the high-minus-low CAR gap is +1.50pp over $[0, +10]$ ($t = 2.23$) and +0.99pp over $[0, +1]$ ($t = 3.72$), and the matched-sample AIIE regression coefficient, +0.76pp ($t = 2.05$), is essentially identical to the full-sample estimate (+0.71pp). The five-day window remains the weak one (an insignificant -0.10pp), exactly as in the main analysis. The priced-exposure result is therefore not an artifact of the size/growth composition of high-AIIE firms; it is present among firms matched on those characteristics.

Randomization inference. Table 10 and Figure 1 report industry-level randomization inference: we permute the 182 industry AIIE values among the industries 1,000 times, holding firms-per-industry fixed, and refit the clustered cross-section. This is the design-honest null—it respects that exposure is an industry attribute—and it is demanding: the actual $[0, +10]$ coefficient (+0.71pp) sits high in the right tail of—but just inside—the 95% permutation interval $[-0.88, +0.79]$, with a randomization p-value of 0.091, while the tight $[0, +1]$ coefficient (+0.35pp) lies right at the interval's upper boundary ($[-0.38, +0.34]$), $p = 0.051$. Read honestly, this *tempers* the headline: the clustered t-statistics (2.5–2.6) imply more precision than the industry-level permutation distribution supports, because there are effectively 182 industry units, not 3,056 firms. The effect is robust at the immediate window and borderline over the diffuse window under the strictest design-based test—evidence of a priced-exposure effect, but not an overwhelming one, and a direct motivation for the sharper, firm-level exposure measure the student will hand-code.

10 Is the Result Robust in Both Directions? Power and Minimum Detectable Effects

Design-based inference cuts both ways: it disciplines the headline, but it also tells us whether the *null* pieces of the paper are informative. Table 11 reports, for each estimate, the minimum detectable effect (MDE) at 80% power, MDE $= (z_{0.975} + z_{0.80})\,\widehat{\mathrm{SE}} \approx 2.80\,\widehat{\mathrm{SE}}$, denominated in percentage points and as a share of the cross-firm CAR standard deviation (CAR means are near zero, so a "percent of mean" scaling would be meaningless). The cross-sectional AIIE regressions are well powered against the long–short magnitudes the market actually delivered: over $[0, +10]$ the design could detect an AIIE effect as small as 0.80pp per standard deviation (about 5.5% of the cross-firm CAR SD), far below the +1.3-to-+2.7pp long–short returns. Note that the per-standard-deviation coefficient itself (+0.71pp) sits just below this 0.80pp detection threshold, which is precisely why the industry-level randomization inference in Section 9 comes back borderline—the two design-based diagnostics corroborate rather than contradict each other. The five-day window's insignificance is therefore an informative bound—an effect there above 0.62pp would likely have shown up—rather than a symptom of an underpowered test, and the placebo window's null (−0.33pp, MDE 0.84pp) is a genuine absence of pre-trend, not noise. The one genuinely imprecise object is the substitution bucket: with only nine substitution industries, the mean $[0, +10]$ CAR of -2.76% carries an MDE of 3.33pp, so while the point estimate is large and negative, the bucket is individually noisy and the sign channel needs the student's larger, hand-coded substitution sample to be pinned down. Stating each null as a bound, rather than as a bare “insignificant,” is what makes the paper’s mixed evidence honest.

11 Conclusion

Within days of ChatGPT’s release, U.S. equity markets repriced firms by their exposure to the cognitive labor generative AI can perform—and did so with a sign that depends on economic role. Firms that build AI, and firms that use exposed labor as an internal input, gained; firms whose product is the very labor AI can now produce lost. A one-SD increase in industry AI exposure is worth roughly +0.7 to +1.0pp of ten-day abnormal return on average, but AI-supplier industries beat substitution-threatened industries by about 2.5pp. The market behaved as the task framework predicts: it priced not just *how exposed* a firm is to AI, but *who captures the surplus.* The priced-exposure result survives characteristic matching—high- and low-AIIE firms balanced on size, value, momentum, and employment still differ by +1.50pp over $[0, +10]$—so it is not a size or growth factor loading in disguise. But industry-level randomization inference, the design-honest test given that AIIE varies across only 182 industries, is more sober than the clustered t-statistics: the headline is robust at the immediate window ($p = 0.051$) and borderline over the diffuse window ($p = 0.091$). A power analysis shows the well-populated cross-sectional windows could have detected effects far smaller than the market delivered, so their nulls (the five-day window, the placebo) are informative bounds; the substitution bucket, by contrast, is genuinely noisy with only nine industries. Limitations include the coarse NAICS-based sign classification, the diffuse and confounded event windows, and the industry-level nature of the exposure measure; the student’s hand-verified event calendar and firm-level task coding are designed to address exactly these, and should both sharpen the sign channel and firm up the borderline diffuse-window inference. Extending to later model releases and to realized earnings would test whether the repricing was justified.

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Table 1: Summary statistics

Variable	N	Mean	SD	P25	Median	P75
CAR[0,+5] ChatGPT	3015	-0.0044	0.1168	-0.0441	-0.0026	0.0339
CAR[0,+10] ChatGPT	3015	-0.0115	0.1727	-0.0712	-0.0118	0.0430
CAR[0,+5] GPT-4	2966	-0.0285	0.1164	-0.0681	-0.0203	0.0122
AIIE (industry AI exposure)	3056	0.6509	0.9748	0.2091	0.4637	1.6354
ln(ME)	3055	13.2971	2.4556	11.4092	13.3132	15.0172
6-mo momentum	3056	-0.0887	0.4186	-0.3239	-0.0942	0.0905
Book/Market	2931	0.8307	1.7970	0.2109	0.4611	0.9860
ln(employees)	2894	-0.1838	2.5053	-2.1037	-0.1393	1.7228

Notes: 3,056 firms. CARs are market-model cumulative abnormal returns around ChatGPT (2022-11-30) and GPT-4 (2023-03-14). AIIE is the Felten–Raj–Seamans AI Industry Exposure (4-digit NAICS). Controls are predetermined (measured before ChatGPT).

Table 2: AI-exposure quintile portfolios: mean CARs and long–short

	Mean CAR by AIIE quintile (%)					Long–short		
Event window	Q1 (low)	Q2	Q3	Q4	Q5 (high)	Q5−Q1	t	N
ChatGPT $[0, +1]$	-1.01	-0.63	-1.34	-0.40	0.31	1.31***	(3.96)	3015
ChatGPT $[0, +5]$	-0.82	0.12	-2.37	-0.22	-0.12	0.70	(1.43)	3015
ChatGPT $[0, +10]$	-1.98	-0.77	-4.23	-0.90	0.70	2.68***	(3.46)	3015
ChatGPT $[-10, -2]$ plac.	-2.11	-2.01	-3.33	-2.75	-2.60	-0.49	(-0.64)	3020
GPT-4 $[0, +5]$	-3.23	-3.97	-3.08	-2.03	-1.33	1.91***	(3.83)	2966
GPT-4 $[0, +10]$	-3.00	-4.58	-3.55	-2.94	-2.07	0.93	(1.43)	2966

Notes: Equal-weighted mean market-model CARs (%) by AIIE quintile, and the top-minus-bottom quintile long–short with a two-sample t. $[-10, -2]$ is a pre-event placebo. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Cross-sectional CAR regressions around ChatGPT

	(1)	(2)	(3)	(4)	(5)
AI Industry Exposure (AIIE)	0.0058*	0.0059**	0.0071**	0.0097***	0.0035***
	(1.88)	(2.07)	(2.49)	(3.39)	(2.61)
ln(ME)		0.0126***	0.0055	0.0065	0.0043
		(3.40)	(0.71)	(0.82)	(1.46)
6-mo momentum		-0.0108*	-0.0092*	-0.0089*	-0.0047***
		(-1.96)	(-1.92)	(-1.92)	(-3.00)
Book/Market			-0.0054	-0.0065	-0.0013
			(-1.36)	(-1.58)	(-0.72)
ln(employees)			0.0052	0.0061	0.0001
			(0.83)	(0.93)	(0.05)
Event window	[0,+10]	[0,+10]	[0,+10]	[0,+10]	[0,+1]
Sample	All	All	All	Non-fin.	All
Industry clusters	180	180	179	178	179
Observations	3015	3014	2864	2711	2864
R^2	0.002	0.010	0.010	0.014	0.013

Notes: DV is the market-model CAR over the stated window. All regressors winsorized 1/99 and standardized (coefficients are the effect of a one-SD increase). Column (4) drops financial (SIC 6000–6999) firms. Standard errors clustered by 4-digit NAICS industry; t-statistics in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: The sign channel: mean CAR by AI-role bucket

	AI supplier		AI user		Substitution	
Event window	CAR%	t	CAR%	t	CAR%	t
ChatGPT [0,+5]	0.36	(0.73)	-0.45*	(-1.88)	-2.11***	(-2.73)
ChatGPT [0,+10]	0.64	(0.76)	-1.28***	(-3.67)	-2.76**	(-2.32)
GPT-4 [0,+5]	-1.49***	(-2.62)	-2.96***	(-12.65)	-3.64***	(-3.51)

Notes: Equal-weighted mean CARs (%) with t-statistics for AI-supplier (builds AI), AI-user (complement), and substitution (AI can produce the firm's output) industries. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: The sign channel: bucket-dummy regression (ChatGPT $[0, +10]$)

DV = CAR[0,+10] ChatGPT (%)	Coef. (%)	t
AI supplier (vs. user)	1.41	(0.87)
Substitution (vs. user)	-1.06	(-1.11)
ln(ME)	0.62	(0.80)
6-mo momentum	-0.95**	(-1.97)
Intercept	-1.40***	(-3.53)
Supplier − Substitution	2.47	
Industry clusters	179	
Observations	2864	
R^2	0.009	

Notes: DV is CAR$[0, +10]$ around ChatGPT (%). Base category is AI users. Predetermined controls included. Standard errors clustered by industry. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Robustness: the AIIE coefficient across windows, events, and estimators

Specification	AIIE coef. (%)	t	N	R^2
ChatGPT [0,+5], HC1 SE	0.20	(1.26)	2864	0.006
ChatGPT [0,+5], drop financials	0.23	(0.97)	2711	0.007
ChatGPT [0,+1]	0.35***	(2.61)	2864	0.013
ChatGPT [-1,+1]	0.23*	(1.68)	2864	0.019
ChatGPT [-10,-2] PLACEBO	-0.33	(-1.11)	2869	0.013
GPT-4 [0,+5] (SVB-contaminated)	0.63***	(2.99)	2822	0.026
GPT-4 [0,+5], drop financials	0.62***	(2.72)	2670	0.027

Notes: Each row is a separate regression of the stated CAR on standardized AIIE with the full control set; coefficient and t on AIIE reported. Clustered by industry except the HC1 row. GPT-4's window overlaps the March-2023 banking crisis. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Firm-level validation: 10-K AI-mention intensity

Dependent variable	10-K AI-intensity coef. (%)	t	N	R^2
CAR[0,+10] ChatGPT	1.06	(1.51)	646	0.023
CAR[0,+1] ChatGPT	0.47**	(2.04)	646	0.016

Notes: 10-K AI-mention intensity (log AI terms per 10,000 words), standardized, from the most recent pre-ChatGPT 10-K on SEC EDGAR for 650 firms. Controls for size and momentum; industry-clustered t. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Heterogeneity of the priced-AIIE effect by observable moderators

	CAR[0,+10] (%)		CAR[0,+1] (%)	
Moderator M (pre-event)	AIIE	AIIE$\times M$	AIIE	AIIE$\times M$
Large firm (high ln ME)	+0.25	+0.77	-0.04	+0.64***
High book/market	+1.08***	-0.73	+0.45***	-0.20
High momentum	+1.16***	-0.87*	+0.43**	-0.16
Financial firm (SIC 6000-6999)	+0.95***	-1.91	+0.38***	-0.17

Notes: Each cell pair comes from one regression of the stated ChatGPT CAR (%) on standardized AIIE, its interaction with the moderator indicator, the moderator, and the full predetermined control set, with industry-clustered standard errors. "AIIE" is the effect in the low-moderator group; "AIIE$\times M$" is the additional effect in the high group. Moderators are predetermined and observable, chosen to be orthogonal to the hand-coded firm-level task exposure reserved for the student. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: Characteristic-matched design: does the priced-AIIE gap survive balancing?

	High−low AIIE gap		Matched regression		
Outcome (DV)	Diff. (%)	t	AIIE (%)	t	N
CAR[0,+10] ChatGPT	+1.50**	(+2.23)	+0.76**	(+2.05)	1350
CAR[0,+1] ChatGPT	+0.99***	(+3.72)	+0.45***	(+2.89)	1350
CAR[0,+5] ChatGPT	-0.10	(-0.23)	+0.13	(+0.52)	1350
Covariate balance (high−low mean gap, standardized):					
	Pre-match	Post-match			
ln(ME)	-0.190	+0.019			
6-mo momentum	-0.146	-0.020			
Book/Market	+0.151	-0.002			
ln(employees)	-0.375	-0.043			

Notes: High-AIIE (top tercile) firms are nearest-neighbor matched without replacement to low-AIIE (bottom tercile) firms on the standardized predetermined controls (size, book-to-market, momentum, employment), Euclidean distance, caliper 0.5. "High−low AIIE gap" is the difference in mean CAR (%) on the matched sample with a two-sample t; "Matched regression" re-estimates the full-control AIIE cross-section on the matched sample (coefficient in %, industry-clustered t). Balance rows report the standardized high-minus-low mean gap before and after matching. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: Industry-level randomization inference

Outcome (DV)	Actual AIIE (%)	Perm. 95% interval (%)	RI p-value
CAR[0,+10] ChatGPT	+0.71	[-0.88, +0.79]	0.091
CAR[0,+1] ChatGPT	+0.35	[-0.38, +0.34]	0.051

Notes: The 182 industry AIIE values are permuted among the industries 1,000 times (firms-per-industry held fixed) and broadcast back to firms; the full-control AIIE cross-section is refit each time. "Actual AIIE" is the estimated coefficient (%); the permutation interval is the central 95% of the permutation distribution; the RI p-value is the share of permutations with |coefficient| at least as large as the actual.

Table 11: Power and minimum detectable effects (80% power)

Estimate	Coef. (pp)	SE (pp)	MDE(80%) (pp)	% of CAR SD
AIIE coef., CAR[0,+1] ChatGPT	+0.35	0.134	0.37	6.8%
AIIE coef., CAR[0,+5] ChatGPT	+0.20	0.220	0.62	6.4%
AIIE coef., CAR[0,+10] ChatGPT	+0.71	0.287	0.80	5.5%
AIIE coef., CAR[-10,-2] placebo	-0.33	0.298	0.84	7.3%
AIIE coef., CAR[0,+5] GPT-4	+0.63	0.212	0.59	5.9%
Substitution bucket mean, CAR[0,+5] ChatGPT	-2.11	0.771	2.16	22.3%
Substitution bucket mean, CAR[0,+10] ChatGPT	-2.76	1.189	3.33	22.7%

Notes: MDE $= (z_{0.975} + z_{0.80})\,\widehat{\mathrm{SE}} \approx 2.80\,\widehat{\mathrm{SE}}$, in percentage points and as a share of the cross-firm CAR standard deviation. AIIE-coefficient rows are the full-control cross-section for each window; substitution-bucket rows are the mean CAR of the nine substitution industries. CAR means are near zero, so effects are not scaled by the mean.

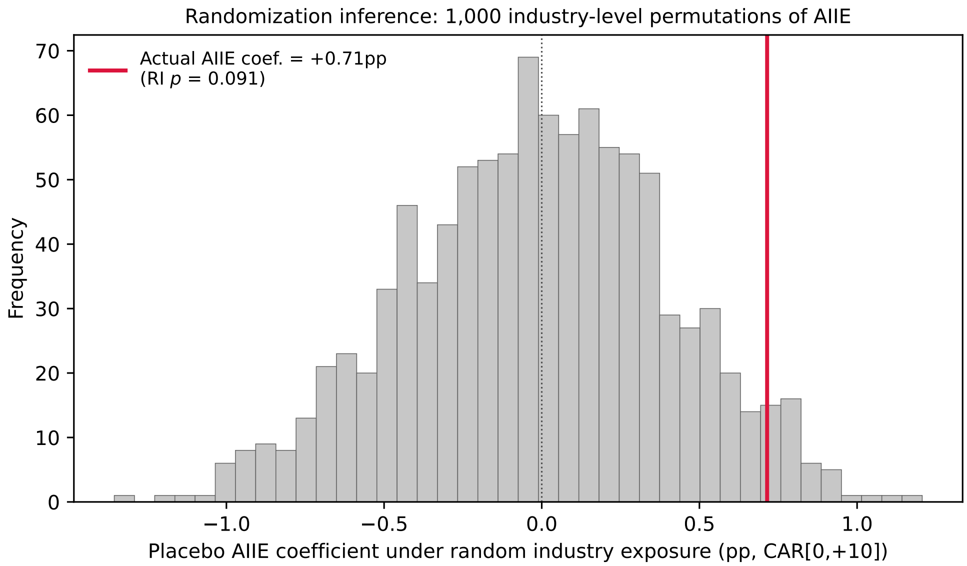

Figure 1: Industry-level randomization inference for the headline window. Histogram of the AIIE coefficient on CAR$[0, +10]$ under 1,000 permutations of industry exposure among the 182 industries; the red line is the actual estimate. The actual coefficient sits in the right tail of the permutation distribution (RI $p = 0.091$), so the design-based test supports—but does not overwhelm—the clustered-SE inference.

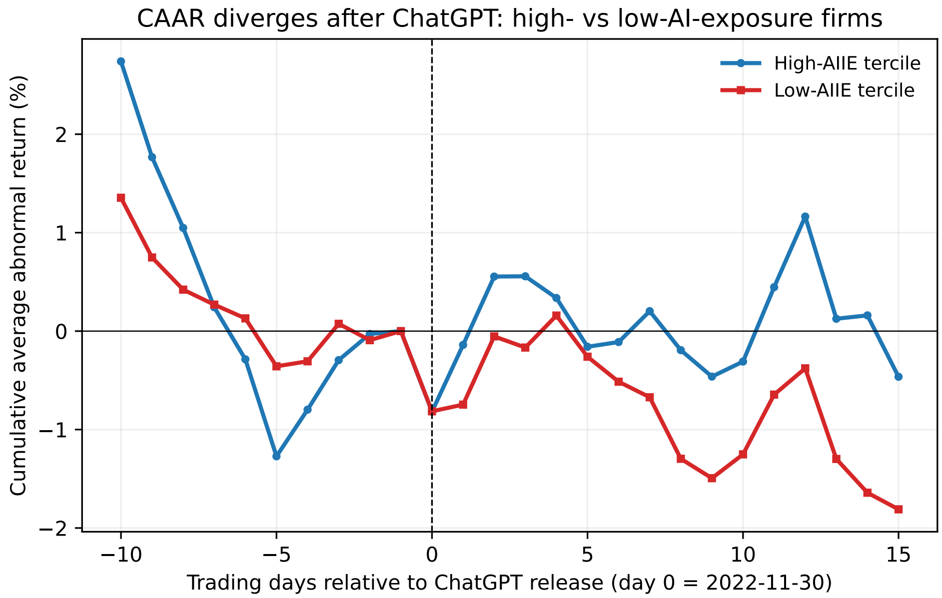

Figure 2: Cumulative average abnormal return (CAAR) around ChatGPT for the top and bottom AIIE terciles, re-based to zero at day -1. The series track each other before the event and diverge afterward.

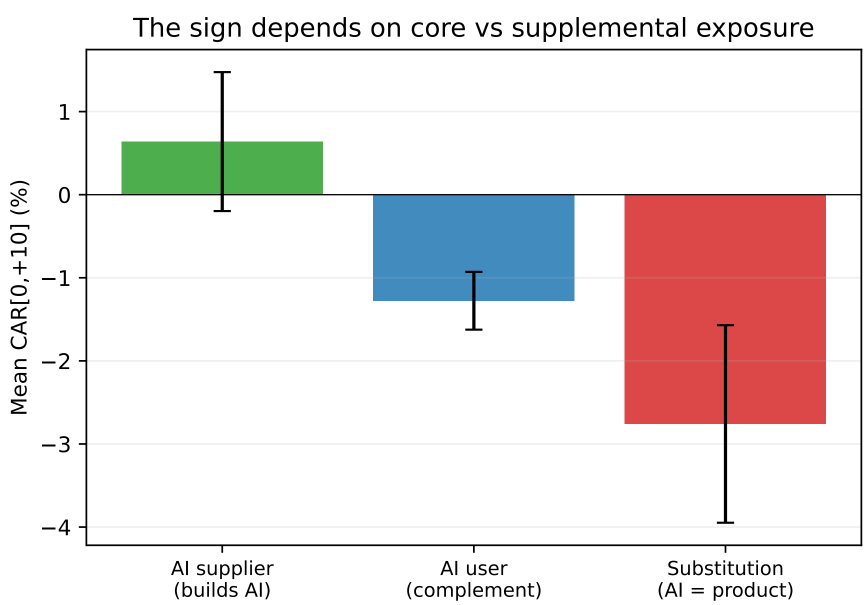

Figure 3: Mean ChatGPT CAR$[0, +10]$ by AI-role bucket. AI suppliers gain; substitution industries (AI can produce their output) lose most. Error bars are ± 1 standard error.

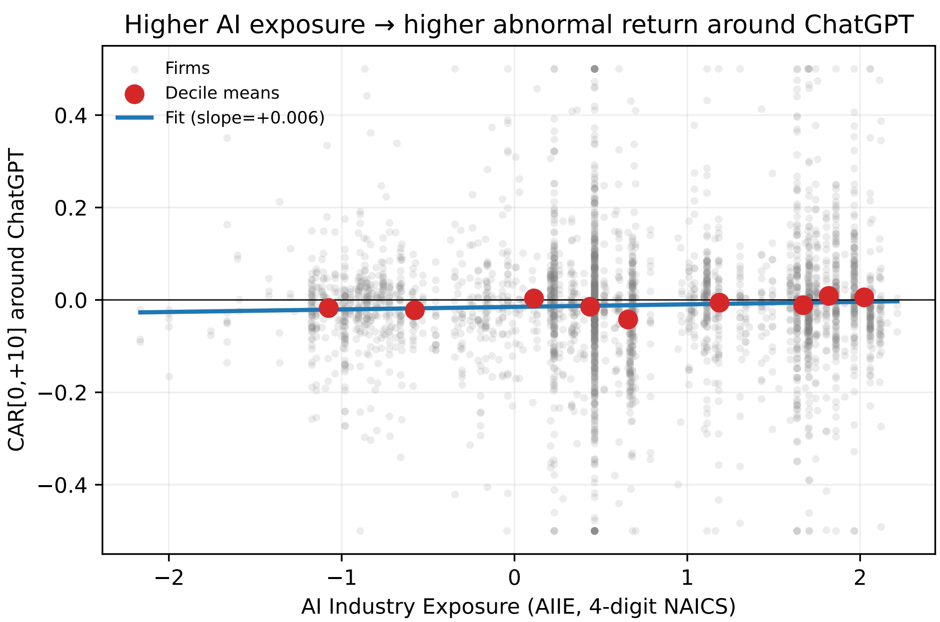

Figure 4: ChatGPT CAR$[0, +10]$ versus AI Industry Exposure. Gray points are firms; red points are AIIE decile means; the line is the OLS fit.